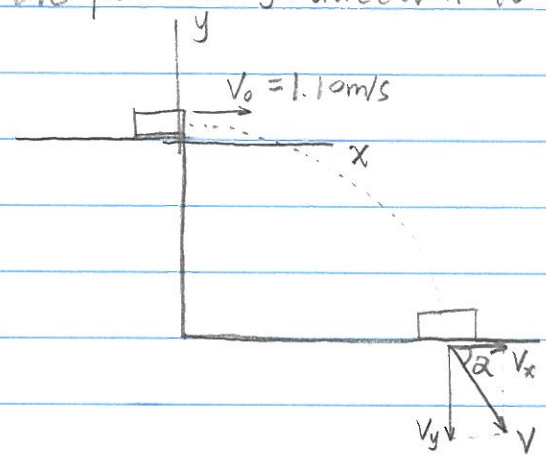
Phys. 2110 Solution

HW # 3 3.9, 26, 52, 56, 64, 77, 82

3.9 Take the positive y-direction to be upward



Take the origin of coordinates at the initial position of the book, at the point where it leaves the tabletop.

(a) $y - y_0 = V_{0y} t + \frac{1}{2} a_y t^2 = 0 + \frac{1}{2} \times (-9.80 m/s^2) \times (0.350 s)^2 = -0.600 m$

The tabletop is 0.600 m above the floor.

(b) $x - x_0 = V_{0x} t + \frac{1}{2} a_x t^2 = 1.10 m/s \times 0.350 s = 0.385 m$

(c) $V_x = 1.10 m/s$

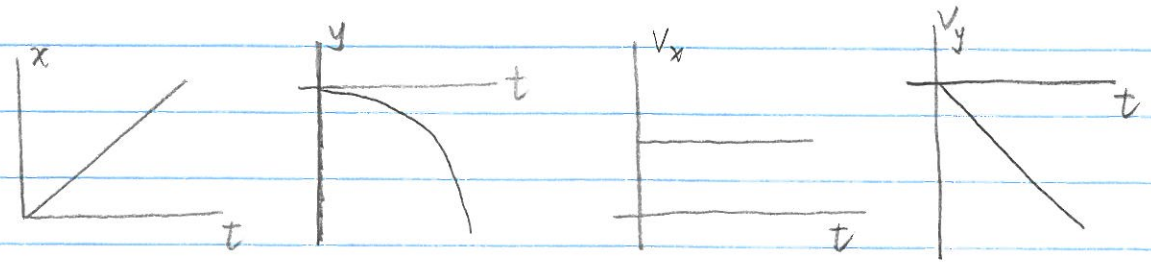
$V_y = V_{0y} + a_y t = 0 + (-9.80 m/s^2) \times (0.350 s) = -3.43 m/s$

$V = \sqrt{V_x^2 + V_y^2} = 3.60 m/s$

$\tan\alpha = \frac{V_y}{V_x} = \frac{-3.43 m/s}{1.10 m/s} = -3.118 \qquad \alpha = -72.2°$

Direction of $\vec{V}$ is 72.2° below the horizontal

(d)



3.26 Let $+y$ be upward $a_x = 0$, $a_y = -g$. $V_{0x} = V_0 \cos 43°$, $V_{0y} = V_0 \sin 43°$

(a) horizontal motion: $x - x_0 = V_{0x} t$, $t = \dfrac{60.0\,m}{V_0 \cos 43°}$

vertical motion: $y - y_0 = V_{0y} t + \frac{1}{2} a_y t^2$

$$25.0\,m = (V_0 \sin 43°) t + \frac{1}{2}(-9.80\,m/s^2) t^2$$

Solving these two equations gives $\boxed{V_0 = 32.6\,m/s}$, $\boxed{t = 2.51\,s}$

(b) V_y when the shell reaches cliff

$$V_y = V_{0y} + a_y t = (32.6\,m/s) \sin 43.0° - 9.80\,m/s^2 \times 2.51\,s$$
$$= -2.4\,m/s$$

The shell is traveling downward when it reaches the cliff, so it lands right at the edge of the cliff.

3.52 (a) Use the vertical motion to find the time in the air

$$y - y_0 = V_{0y} t + \frac{1}{2} a_y t^2, \quad y - y_0 = -30.0\,m$$

$$-30.0\,m = (10.0\,m/s) t + \frac{1}{2}(-9.80\,m/s^2) t^2$$

gives $$t = \frac{1}{2 \times 4.9}\left(10.0 \pm \sqrt{(-10.0)^2 - 4 \times 4.9 \times (-30)}\right) s$$

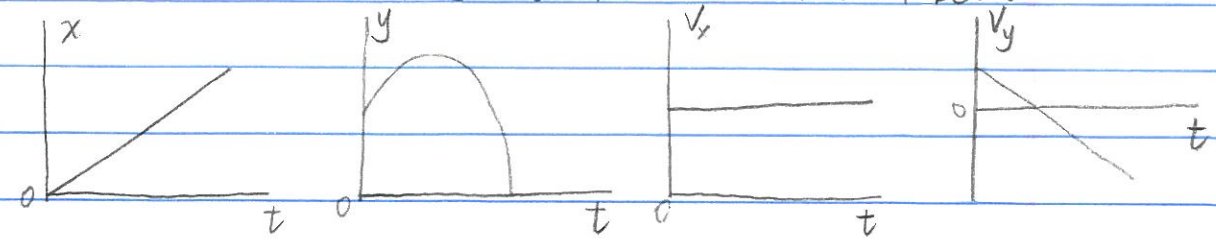
the positive solution is $t = 3.70\,s$

During this time she travels a horizontal distance

$$x - x_0 = V_{0x} t + \frac{1}{2} a_x t^2 = 15.0\,m/s \times 3.70\,s = 55.5\,m$$

She will land 55.5m south of the point where she drops from the helicopter.

(b) The x-t, y-t, V_x-t, V_y-t graphs are sketched below



3.56 The equations of motions are $x = (V_0 \cos\alpha)t$

$y = (V_0 \sin\alpha)t - \frac{1}{2}gt^2$, $\sin\alpha = \cos\alpha = \frac{\sqrt{2}}{2}$

(i) When the water enters the tank for the minimum velocity

$y = 2D$, $x = 6D$

$x = (V_0 \cos\alpha)t \qquad 6D = \frac{\sqrt{2}}{2} V_0 t \quad \Rightarrow \quad t = \frac{6\sqrt{2}D}{V_0}$

$y = (V_0 \sin\alpha)t - \frac{1}{2}gt^2 \qquad 2D = \frac{\sqrt{2}}{2} V_0 t - \frac{1}{2}gt^2$

Substitute t, $\quad 2D = 6D - \frac{1}{2}g\left(\frac{6\sqrt{2}D}{V_0}\right)^2$

Solving this for V_0 gives $\boxed{V_0 = 3\sqrt{gD}}$

(ii) When the water enters the tank for the maximum velocity

$y = 2D$, $x = 7D$

$7D = \frac{\sqrt{2}}{2} V_0 t \quad \Rightarrow \quad t = \frac{7\sqrt{2}D}{V_0}$

$2D = \frac{\sqrt{2}}{2} V_0 t - \frac{1}{2}gt^2 = 7D - \frac{1}{2}g\left(\frac{7\sqrt{2}D}{V_0}\right)^2$

Solving this for V_0 gives $V_0 = \sqrt{\frac{49}{5}gD} = \boxed{3.13\sqrt{gD}}$

So the range of lauch speeds V_0 is $3\sqrt{gD}$ — $3.13\sqrt{gD}$

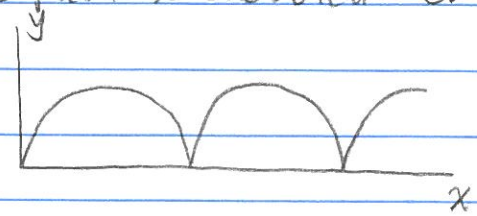
3.64 projectile motions $\quad V_x = V_0 \cos\alpha_0$, $V_y = V_0 \sin\alpha_0 - gt$

$V^2 = V_x^2 + V_y^2 = V_0^2 \cos^2\alpha_0 + (V_0 \sin\alpha_0 - gt)^2 = V_0^2 - 2g(V_0 \sin\alpha_0 t - \frac{1}{2}gt^2)$

$V^2 = V_0^2 - 2gy$, $\quad y = (V_0 \sin\alpha_0)t - \frac{1}{2}gt^2$

For the case of a rock thrown from the roof of a building of height h, the speed at the ground is found by substituting $y = -h$ into the above expression, yielding $V = \sqrt{V_0^2 + 2gh}$, which is independent of α_0

3.77 (a) The path is sketched below



(b) velocity components

$$V_x = \frac{dx}{dt} = R\omega(1-\cos\omega t) \qquad V_y = \frac{dy}{dt} = R\omega \sin\omega t$$

acceleration components

$$a_x = \frac{dV_x}{dt} = R\omega^2 \sin\omega t \qquad a_y = \frac{dV_y}{dt} = R\omega^2 \cos\omega t$$

(c) The particle is at rest ($V_y = V_x = 0$) every period, at $t=0, 2\pi/\omega, 4\pi/\omega, \cdots$ At that time, $x = 0, 2\pi R, 4\pi R, \cdots$ and $y=0$.

The acceleration is $a = R\omega^2$ in the $+y$ direction.

(d) No. since $a = [(R\omega^2 \sin\omega t)^2 + (R\omega^2 \cos\omega t)^2]^{1/2} = R\omega^2$.

The magnitude of the acceleration is the same as for uniform circular motion.

3.82 (a) $-3.00\,m = -\frac{1}{2} \times (9.80\,m/s^2)\,t^2 \Rightarrow \boxed{t = 0.782\,s}$

(b) Relative to an observer in the elevator, the velocity of the bolt is $-(9.80\,m/s^2) \times (0.782\,s) = \boxed{-7.67\,m/s}$

(c) Relative to the Earth, the velocity of the bolt is

$$2.50\,m/s - (9.80\,m/s^2) \times (0.782\,s) = \boxed{-5.17\,m/s}$$

(d) Relative to the Earth, the distance the bolt traveled is

$$(2.50\,m/s)\,t - \frac{1}{2}(9.80\,m/s^2)\,t^2 = (2.50\,m/s) \times (0.782\,s) - \frac{1}{2} \times (9.80\,m/s^2) \times (0.782\,s)^2 = \boxed{-1.04\,m}$$